

Negotiated Affect: Designing Performative Input and Withdrawal for Camera-Based Affective Installations

Zhijun Ma

Computational Media and Arts Thrust, Information Hub
The Hong Kong University of Science and Technology (Guangzhou)
Guangzhou, China
zma361@connect.hkust-gz.edu.cn

Abstract

Camera-based affect sensing in public settings is often interpreted through a "truth detection" lens, raising concerns about surveillance, misrecognition, and visitors' loss of control. We present the performative interface, a generative audiovisual installation deployed in an exhibition space that uses face-derived affect cues to modulate real-time visuals and sound. Rather than treating sensed affect as inner truth, we propose Performative Calibration, an interaction policy that reframes affect sensing as a performable input grammar. The policy is implemented through (1) an intent threshold that privileges deliberate, sustained expressions over incidental micro-expressions, (2) a diegetic confirmation loop that acknowledges input without textual UI, (3) a withdrawal gesture that lets visitors visibly pause sensing and enter an explicit refusal/blank state. In a qualitative exhibition study, participants shifted from "being read" to "playing an instrument", using exaggeration, masking, and boundary-testing to discover and perform system rules. We discuss implications for designing states in camera-based affective interfaces.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; *Gestural input*; • **Social and professional topics** → Computing / technology policy.

Keywords

Agency, Ethical HCI, Affective sensing, Negotiated interaction, Performativity, Public installations.

ACM Reference Format:

Zhijun Ma. 2026. Negotiated Affect: Designing Performative Input and Withdrawal for Camera-Based Affective Installations. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3798807>

1 Introduction

Camera-based affect sensing in public spaces is frequently perceived as "reading people," which amplifies concerns around surveillance, misrecognition, and loss of control. While affective computing has matured into a broad set of models and applications [1, 4, 11],

face-derived emotion signals remain common because cameras are ubiquitous and interaction can be "hands-free" [6, 10]. Yet in exhibitions and other public venues, sensing is socially visible: visitors may feel judged by the system and by bystanders, and may not know what counts as input or how to stop being sensed without leaving.

Much prior discussion focuses on model accuracy, dataset ambiguity, and deployment risks [3, 12]. These issues matter, but they do not fully address a more interactional question: what mechanisms let people negotiate when and how sensing becomes actionable input? In many installations, the camera stream is continuously interpreted, creating "always-on" power asymmetries reminiscent of surveillance framings [3, 6]. In contrast, HCI work on context-aware systems has long questioned whether sensing and automation take control away from users, and calls for legible control at the interaction level [2].

We propose *Performative Calibration*: an interaction policy that treats sensed affect not as inner truth but as a performable input grammar that visitors can deliberately perform, modulate, or refuse. We implement this policy in *the performative interface*, a camera-driven generative audiovisual installation. Performative Calibration is operationalized through (1) an intent threshold that privileges deliberate, sustained expressions; (2) a diegetic confirmation loop that acknowledges input without textual UI; and (3) a withdrawal gesture that pauses sensing and enters a visible refusal/blank state.

We contribute:

- **Performative Calibration**, an interaction policy for camera-based affective installations that makes withdrawal a first-class, reversible interaction state.
- **An exhibition qualitative study** (N=8) showing how visitors develop negotiation strategies (exaggeration, holding, masking, boundary-testing) and how withdrawal changes comfort and participation.
- **Design implications** for agency, consent-in-interaction, and passive/active withdrawal in public affective interfaces.

We motivate a shift from accuracy-only debates toward interaction mechanisms that support performable input and reversible refusal in public affect sensing.

2 Related Work

2.1 Affective Sensing, Surveillance Framings, and Misrecognition

Affective computing integrates psychological models and computational methods [4], with recent surveys highlighting learning-based multimodal pipelines [1, 11]. However, public-facing affect



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798807>

sensing can be interpreted as surveillance-like "truth detection," a concern raised early in discussions of emotion recognition as biometric surveillance [3] and reiterated in contemporary facial emotion recognition work [6]. Technical limitations compound these risks: emotion labels can be ambiguous and inconsistently annotated, which affects model behavior and perceived legitimacy [12]. Privacy-preserving techniques (e.g., processing in encrypted domains) can reduce raw data exposure [9], but do not by themselves provide in-the-moment interaction control.

2.2 Agency, Performativity, and Legible Control

HCI and ubiquitous computing research has emphasized that sensing systems should not quietly remove user control; instead, interaction should clarify what the system is doing and what the user can do about it [2]. In public interactive settings, engagement is often performative: people exaggerate, improvise, and treat the system as something to be "played." Affective interfaces can support such experiences (e.g., responding to frustration) [7], but may also present an illusion of empathy that misleads users about what the system understands or intends [5]. This motivates designs that surface a controllable grammar rather than implying emotional truth.

2.3 Withdrawal, Opt-Out, and "Refusal as Interaction"

Refusal is frequently treated as policy or signage rather than interaction. Yet in sensitive contexts, explicit controls (e.g., "panic buttons") can create empowerment by enabling immediate disengagement [8]. For public affect sensing, a key challenge is making refusal both *legible* (the system shows it is not sensing) and *reversible* (people can rejoin without friction). We extend this line by treating withdrawal as a designed state that supports safe exploration and boundary-setting in camera-based affective installations.

Prior work establishes ethical concerns in affect sensing, the importance of legible control in sensing systems, and the value of explicit disengagement mechanisms. We focus on an interaction policy that makes negotiation and withdrawal practical, visible, and reversible.

3 System: The Performative Interface

3.1 Overview and Focus

The *performative interface* is a camera-driven generative audiovisual installation deployed in an exhibition space. A visitor's face-derived affect cues modulate real-time visuals and sound to evoke coarse scenes (calm, storm, mist). This paper focuses on the interaction policy that mediates sensing into input and refusal, rather than on visual style or claims of emotion "ground truth" [4].

3.2 Pipeline and Policy Layer

Figure 1 summarizes the pipeline. A camera captures frames; a face region is extracted; an affect model outputs a streaming signal (labels/scores); a policy layer determines whether the signal becomes control input, triggers acknowledgment, or is ignored. Placing policy between sensing and output is intentional: it allows ethical/agency constraints to shape interaction even when model outputs are noisy or ambiguous [12].

3.3 Performative Calibration

Performative Calibration reframes sensed affect as a *performable input grammar*. Visitors can learn what counts as input, deliberately provide it, and withdraw without leaving. Figure 2 summarizes the three mechanisms and their state transitions from affect stream to registered input and refusal.

3.3.1 Intent Threshold: Prefer Deliberate Expressions. The *intent threshold* registers an input only when the affect signal appears deliberate rather than incidental. Concretely, the system requires:

- **Dwell:** a stable dominant label for at least T_d seconds;
- **Strength:** confidence/score above threshold τ for most frames in the window.

In deployment we used $T_d \approx 1.0s$ and a conservative τ tuned to minimize accidental triggers. This supports *passive withdrawal*: if a visitor does not perform sustained expressions, the system largely does not react, reducing the sense of being continuously "read."

3.3.2 Diegetic Confirmation Loop: Acknowledge Without Text. When the intent threshold is met, the system plays a brief *diegetic confirmation* motif (a short visual "pulse" and sound cue) to acknowledge that input was registered. This avoids instructional UI that can feel policing in public spaces, and helps visitors iteratively learn the grammar through feedback. It also mitigates the risk of implying empathic understanding: the system acknowledges *registration* rather than claiming emotional insight [5].

3.3.3 Withdrawal Gesture: Visible, Reversible Refusal. A *withdrawal gesture* (covering the face, turning away, or stepping out of the sensing zone) immediately pauses sensing and transitions the display into a visible blank/refusal state (muted visuals and subdued audio). This is designed as a dignified mode, not an error. Re-entry is immediate and reversible: returning to the zone resumes idle mode.

Public affect sensing is often criticized as surveillance-like because participation is hard to refuse in the moment [3, 6]. Making withdrawal explicit and reversible parallels empowerment patterns such as panic buttons in sensitive services [8], adapted here for camera-based, non-contact interaction.

The system implements three lightweight mechanisms that collectively shift public affect sensing from "always-on inference" to negotiated, performable input with an explicit reversible refusal state.

4 Method: Exhibition Qualitative Study

We conducted an exploratory, small-scale qualitative study ($N = 8$) during an exhibition deployment to understand how visitors experience this performable affect sensing and how they use (or ignore) withdrawal and calibration mechanisms in practice.

4.1 Setting and Participants

The installation was deployed in a gallery-like room with controlled lighting and a clearly marked interaction zone (approximately 2–3 meters from the display). Recruitment was walk-up and on-site; visitors were invited to participate in a short study session. We report data from $N=8$ participants who consented to observation and a short interview. Participants represented a mix of ages (early

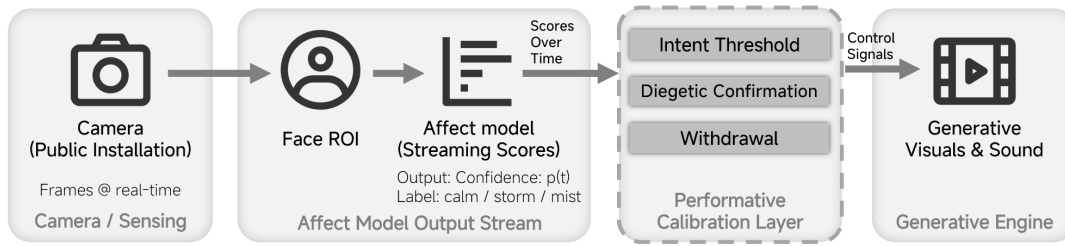


Figure 1: System pipeline and the Performative Calibration policy layer.

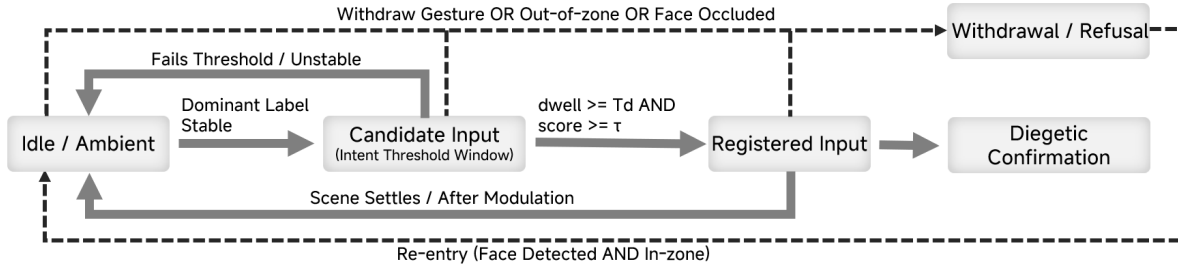


Figure 2: Performative Calibration as an interaction policy. The intent threshold reduces incidental triggers, diegetic confirmation acknowledges registered input without textual UI, and withdrawal enables a visible, reversible refusal state.

20s to mid 40s) and genders; detailed demographics are omitted to preserve anonymity in the exhibition context.

We obtained informed consent for video recording and clarified that the system used camera-based sensing to drive generative audiovisual output. We also explained that the paper focuses on interaction and agency rather than emotion "truth" or diagnosing internal states.

4.2 Procedure

We framed the session as playful exploration rather than testing "true emotion." Participants were invited to "play" three scenes (calm/storm/mist) in any order, and were explicitly told that pausing or withdrawing was allowed. This prompt encouraged both experimentation and boundary-setting.

4.3 Data and Analysis

We collected field notes, video of interaction and screen output (faces blurred in storage), and semi-structured interviews focused on control, comfort, and refusal. We performed an inductive thematic analysis on interview transcripts to identify experiential shifts, alongside light behavior coding to categorize physical interaction strategies (holding, exaggeration, masking, boundary-testing, withdrawal).

The study uses an exhibition-based qualitative approach to observe how people negotiate camera-based affect input when withdrawal and deliberate triggering are designed into the interaction.

5 Findings

Across sessions, participants treated the system less as an evaluator of their "true emotion" and more as an instrument with learnable

interaction rules. We report four themes that describe how Performative Calibration shaped experience and behavior.

5.1 Role Shift: From "Being Read" to "Playing"

Participants often began with caution, describing the camera as evaluative. After experiencing confirmation motifs and learning that brief expressions did not immediately trigger changes, many reframed interaction as "playing" an instrument. This reframing reduced anxiety about being judged and increased exploratory engagement.

5.2 Negotiation Strategies: Learning the Grammar

Visitors developed strategies to reliably register input: exaggerating expressions, holding them until confirmation occurred, repeating inputs to test responsiveness, and occasionally masking parts of the face to probe boundaries. Rather than treating misrecognitions as purely failures, participants frequently treated them as part of discovering the system's rules—a stance enabled by the system's non-"truth" framing [4].

5.3 Withdrawal as Relief and Boundary-Setting

The withdrawal gesture was commonly used as a relief valve when participants felt self-conscious, especially with bystanders present. The visible blank/refusal state made the pause feel legitimate and socially comprehensible. Participants described withdrawal as making the experience "safer to try things," because they could quickly stop sensing without leaving the interaction area—a dynamic consistent with prior work where explicit exit controls support empowerment [8].

5.4 Passive Withdrawal: Spectating Without Triggers

Some participants preferred to spectate or minimally perform. The intent threshold supported this by preventing incidental micro-expressions from continuously driving the installation. Participants reported appreciating that the system did not react to "every tiny thing," suggesting non-performance can be a valid participation mode in public installations [2].

Performative Calibration supported a shift toward instrumental play, produced a learnable negotiation grammar, and made both active and passive withdrawal practical and meaningful.

6 Discussion and Implications

Our findings suggest that ethical progress for public affect sensing is not only about improving recognition models, but also about designing interaction policies that re-balance experiential agency in situ. Importantly, we distinguish between experiential agency and structural surveillance: while withdrawal pauses the system's response and creates a meaningful shift from "being read" to "playing", it does not eliminate the underlying sensing infrastructure. The rebalancing of power is therefore interactional rather than structural. In surveillance framings, the harm is partly that sensing is unilateral and difficult to contest [3, 6]. Performative Calibration offers a different contract: affect cues become optional, performable inputs with explicit refusal.

6.1 Design Implications

- **Design refusal as a visible, dignified state.** A blank/refusal mode communicates "not sensing" without requiring signage interpretation or exiting the space.
- **Make refusal reversible.** Reversible withdrawal lowers the social cost of opting out and encourages safe exploration.
- **Support passive withdrawal via intent thresholds.** "Not performing" should be a legitimate mode, enabling spectating without continuous interpretation [2].
- **Use diegetic acknowledgment to avoid UI policing.** Confirming registration without text reduces the sense of compliance checking and avoids overstating empathic understanding [5].

6.2 Scope and Ethics Position

We do not claim to infer emotional truth, nor to solve dataset ambiguity or fairness problems [12]. Instead, we contribute an interaction policy that reduces incidental interpretation and makes consent-in-interaction more actionable. Technical privacy protections (encrypted-domain processing) can complement this policy [9], but do not replace the need for legible, user-controlled withdrawal in public spaces.

Treating negotiation and withdrawal as interaction primitives offers a pragmatic route to agency in public affective installations, without relying on strong claims about emotion inference.

7 Limitations

This study is small (N=8) and situated in an exhibition context; novelty and audience expectations may affect behavior. Withdrawal

gestures may have cultural and accessibility differences, and we did not compare alternative gesture designs or thresholds. Future work should test variations across contexts (e.g., busier public spaces), examine longer-term use, and evaluate inclusive withdrawal options.

The results demonstrate feasibility and qualitative value, but broader deployments and inclusive gesture design remain necessary.

8 Conclusion

We presented Performative Calibration, an interaction policy for camera-based affective installations that reframes sensed affect as negotiable input rather than truth inference. Through an intent threshold, diegetic confirmation, and a reversible withdrawal gesture, visitors could play the system deliberately, spectate without triggers, and refuse sensing without leaving. We argue that reversible refusal states and performable input grammars should be treated as core design requirements for ethical public affect sensing.

References

- [1] Naveed Ahmed, Zaher Al Aghbari, and Shini Girija. 2023. A Systematic Survey on Multimodal Emotion Recognition Using Learning Algorithms. *INTELLIGENT SYSTEMS WITH APPLICATIONS* (2023).
- [2] Louise Barkhuus and Anind Dey. 2003. Is context-aware computing taking control away from the user? Three levels of interactivity examined. In *International Conference on Ubiquitous Computing*. Springer, 149–156.
- [3] Joseph Bullington. 2005. 'Affective' computing and emotion recognition systems: The future of biometric surveillance?. In *Proceedings of the 2nd annual conference on Information security curriculum development*. 95–99.
- [4] Rafael A. Calvo and Sidney D'Mello. 2010. Affect Detection: an Interdisciplinary Review of Models, Methods, and Their Applications. *IEEE TRANSACTIONS ON AFFECTIVE COMPUTING* (2010).
- [5] Andrea Cuadra, Maria Wang, Lynn Andrea Stein, Malte F Jung, Nicola Dell, Deborah Estrin, and James A Landay. 2024. The illusion of empathy? notes on displays of emotion in human-computer interaction. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [6] Vishnu Kant, Panchal Sandeep Govindrao, and Shubham Mahajan. 2024. Facial Emotion Recognition Using CNNs: Implications for Affective Computing and Surveillance. In *2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES)*.
- [7] Jonathan Klein, Youngme Moon, and Rosalind W Picard. 1999. This computer responds to user frustration. In *CHI'99 extended abstracts on Human factors in computing systems*. 242–243.
- [8] Sofia Lundmark and Ann-Carita Evaldsson. 2016. Click-guides and Panic Buttons: Designed Possibilities for Youth Agency and User Empowerment in Online Youth Counselling Services. *Childhood* (2016).
- [9] Yogachandran Rahulamathavan, Raphael C.-W. Phan, Jonathon A. Chambers, and David J. Parish. 2013. Facial Expression Recognition in the Encrypted Domain Based on Local Fisher Discriminant Analysis. *IEEE TRANSACTIONS ON AFFECTIVE COMPUTING* (2013).
- [10] Yingli Tian, Takeo Kanade, and Jeffrey F Cohn. 2011. Facial expression recognition. In *Handbook of face recognition*. Springer, 487–519.
- [11] Hongxun Yao, Weihong Deng, Honghai Liu, Xiaopeng Hong, Sujing Wang, Jufeng Yang, and Sicheng Zhao. 2022. An Overview of Research Development of Affective Computing and Understanding. *Journal of Image and Graphics* (2022).
- [12] Jiabei Zeng, Shiguang Shan, and Xilin Chen. 2018. Facial Expression Recognition with Inconsistently Annotated Datasets. In *Computer Vision – ECCV 2018*.